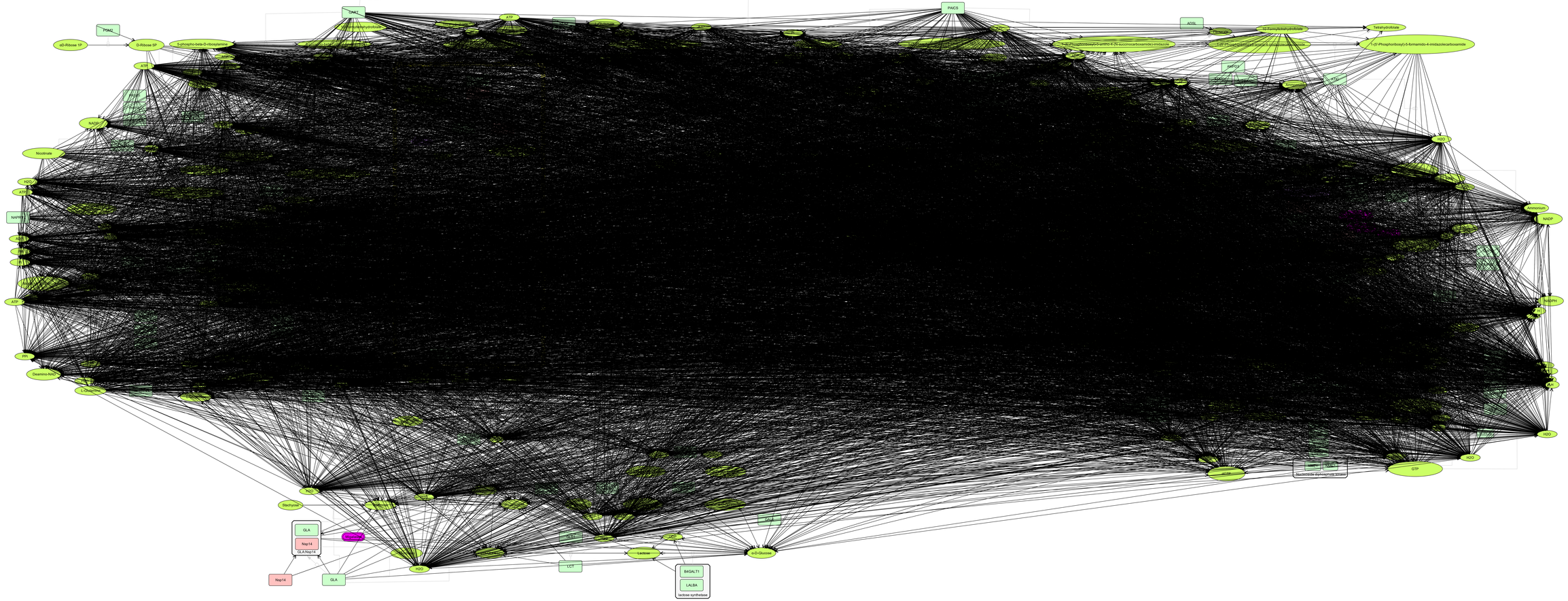

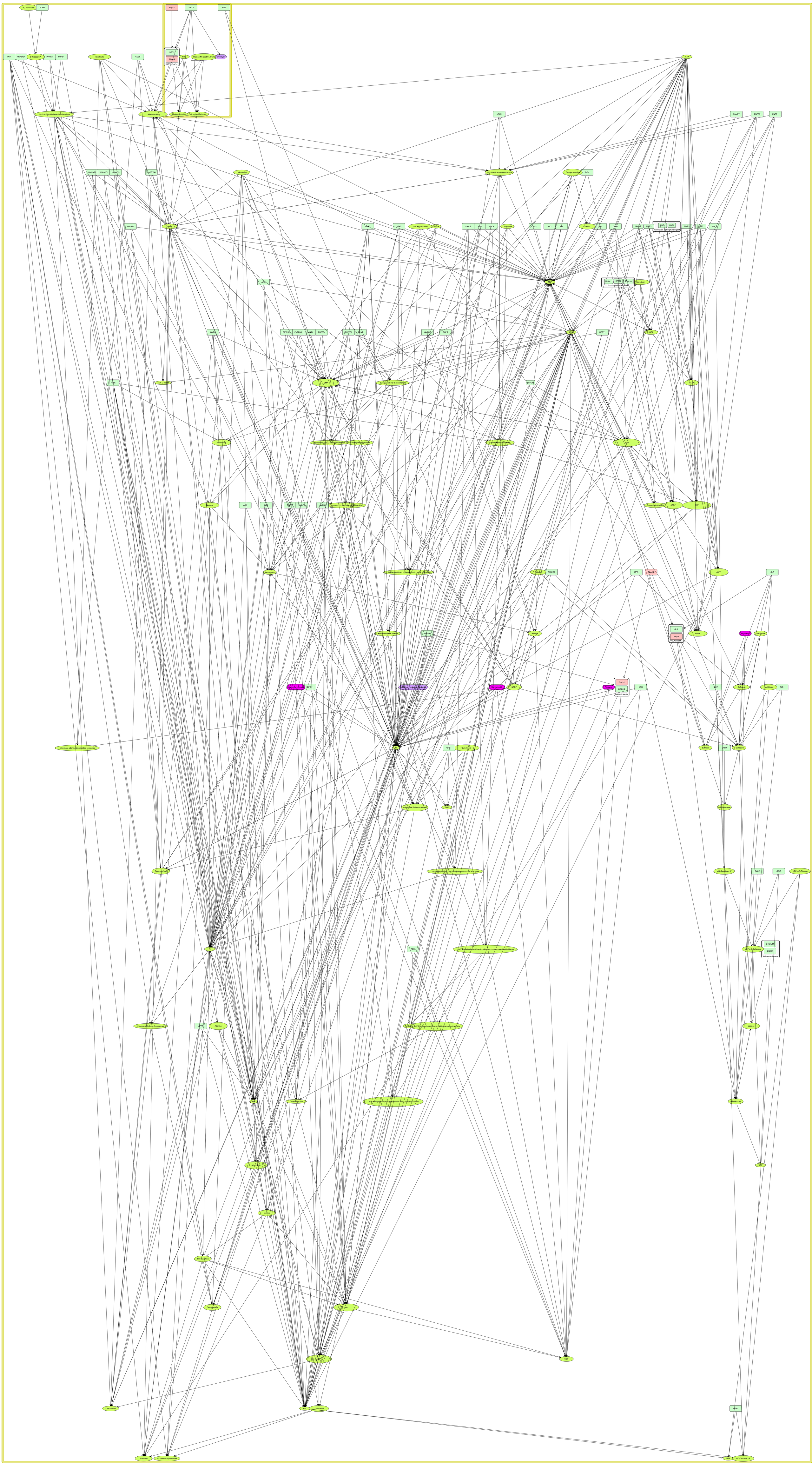

αD-Ribose 1P
PGM2
D-Ribose 5P
5-phospho-beta-D-ribosylamine
GART
10-Formyltetrahydrofolate
ATP
ATP
L-Glutamate
Pi
PAICS
ATP
ADP
ADSL
Fumarate
10-Formyltetrahydrofolate
Tetrahydrofolate
1-(5'-Phosphoribosyl)-5-amino-4-(N-succinocarboxamide)-imidazole
1-(5'-Phosphoribosyl)-5-amino-4-imidazolecarboxamide
1-(5'-Phosphoribosyl)-5-formamido-4-imidazolecarboxamide
AMPD3
AMPD1
AMPD2
ATIC
Ammonium
H2O
NAD
Ammonium
NADP
NADPH
H
H
H2O
H2O
NTPCR
GTP
dGTP
NME1
NME2
Nucleoside diphosphate kinase
ATP
PRPS1
PRPS2
NADP
Nicotinate
H2O
ATP
NAPRT1
ADP
Pi
Pi
ATP
PPi
Deamino-NAD
H2O
L-Glutamine
H2O
Stachyose
Raffinose
H2O
GLA
Nsp14
GLA:Nsp14
Migalastat
Melibiose
H2O
Nsp14
GLA
GLB1
H2O
UDP
Lactose
α-D-Glucose
GALE
LCT
B4GALT1
LALBA
lactose synthetase
cell

αD-Ribose 1P
PGM2
Nsp14
SIRT5
NNT
PNP
PRPS1L1
D-Ribose 5P
PRPS2
PRPS1
Nicotinate
CD38
SIRT5
Nsp14
SIRT5:Nsp14
H2O
Histone N6-acetyl-L-lysine
Urea cycle
mitochondrion
ATP
5-phospho-α-D-ribose 1-diphosphate
Nicotinamide
Histone-L-lysine
O-Acetyl-ADP-ribose
NRK1
NAMPT
ENPP3
ENPP1
NMNAT2
NMNAT1
NMNAT3
NADSYN1
L-Glutamine
Nicotinamide D-ribonucleotide
Deoxyadenosine
DCK
NAPRT1
NAD
GART
PFAS
Deoxyguanosine
Glycine
PAICS
ADK
NADK
L-Aspartate
AK7
AK1
AK8
dAMP
AK5
GUK1
NME6
NME3
NME1
NME2
Nucleoside diphosphate kinase
NME5
NME7
GALK1
ATIC
ADP
RRM1
RRM2
RRM2B
ribonucleoside reductase
Thioredoxin
GMPS
ENTPD5
ENTPD6
CANT1
ENTPD4
ENTPD2
PPAT
GMPR2
GMPR
H2O
HPRT1
dADP
NT5E
ADP-D-ribose
GMP
5-phospho-beta-D-ribosylamine
NTPCR
dATP
Guanosine
5-phospho-beta-D-ribosylglycinamide
10-Formyltetrahydrofolate
N-Ribosyl-nicotinamide
GDP
Guanine
ADA
GDA
AMPD3
AMPD2
AMPD1
5-phosphoribosyl-N-formylglycinamide
Thioredoxin disulfide
dGDP
GTP
Ammonium
2-(Formamido)-N1-(5'-phosphoribosyl)acetamidine
Galactitol
AKR1B1
ITPA
Nsp14
dGTP
GLA
Aminoimidazole ribotide
IMPDH2
NADPH
GLA
Nsp14
GLA:Nsp14
dGMP
Migalastat
Stachyose
Mycophenolic acid
IMPDH1
Guanine nucleotide synthesis
Mizoribine
NADP
Ribavirin
Nsp14
IMPDH2
IMPDH2:Nsp14
XDH
LCT
Raffinose
Melibiose
GLB1
nicotinate-adenine dinucleotide phosphate
H
QPRT
Quinolinate
Sucrose
GALM
D-Galactose
Nicotinate D-ribonucleotide
CO2
α-D-Galactose
Deamino-NAD
1-(5-Phospho-D-ribosyl)-5-amino-4-imidazolecarboxylate
α-D-Galactose-1P
GALE
GALT
UDP-α-D-Glucose
Pi
ADSL
1-(5'-Phosphoribosyl)-5-amino-4-(N-succinocarboxamide)-imidazole
UDP-α-D-Galactose
B4GALT1
LALBA
lactose synthetase
2-deoxy-α-D-ribose 1-phosphate
APRT
Adenine
Fumarate
1-(5'-Phosphoribosyl)-5-amino-4-imidazolecarboxamide
Lactose
AMP
Tetrahydrofolate
1-(5'-Phosphoribosyl)-5-formamido-4-imidazolecarboxamide
α-D-Glucose
Adenosine
UDP
Inosine
Hypoxanthine
NADH
Deoxyinosine
IMP
XMP
UGP2
L-Glutamate
PPi
Xanthosine
UTP
α-D-Glucose-1-P
Xanthine
α-D-Ribose 1-phosphate